

PROGRAMMABLE MONEY · AGENTIC ECONOMY · ARC

Bondwire

Trust and settlement primitives for autonomous agents, settled in USDC on Arc.

LIVE ON ARC TESTNET

CHAIN 5042002

OWNERLESS CONTRACTS

COMMITSTAKEV2 EXACT MATCH VERIFIED

`github.com/Mnorbert87/bondwire`

`mnorbert87.github.io/bondwire`

THE PROBLEM

Agents can already pay. They still cannot be trusted.

Every agentic payment demo quietly assumes two things that do not exist on chain yet.

MISSING ASSUMPTION 1

You can trust the agent before it touches your money

There is no cost to being a new agent. Reputation without stake is a signup form, and a burned identity is replaced for free.

MISSING ASSUMPTION 2

You can pay it while it works, not after

Invoicing assumes a relationship and a clock both parties accept. Autonomous counterparties have neither.

WHERE THE FIELD STOPS

x402 and most of the category solve one act: pay per call. The open question is the one after the payment leaves. **Who pays when the agent defaults, and who pays when the verifier lies?** That is the layer Bondwire ships.

Three ownerless primitives that compose

No owner, no admin role, no proxy, no pause. The only roles are the ones in the protocol itself, scoped per obligation.

TRUST LAYER

AgentBond

An agent deposits USDC as skin in the game. Any enforcer it approves can lock a slice behind an obligation and slash it on default. Free bond is a money backed credit score.

0xB9b4...f8e0

SETTLEMENT LAYER

StreamPay

Continuous USDC settlement. Funds accrue per second, the recipient withdraws any time, either side cancels with a fair split. Pay per second of work, not per invoice.

0x5057...2450

MECHANISM

CommitStakeV2

Pay only on verified PASS. The verifier posts its verdict with its own bond slice locked behind it. A challenge window and an optional arbiter give recourse. Lying costs real money.

0x1f1CA31b...698CA9

WHY COMPOSABLE MATTERS

The two primitives stand alone. Together they form a complete trust and pay rail: a stream can fund a bond, a fulfilled commitment can release one, and a false verdict burns a bonded slice. That composition is only free and permissionless on a shared ledger.

Every claim here is a transaction you can open

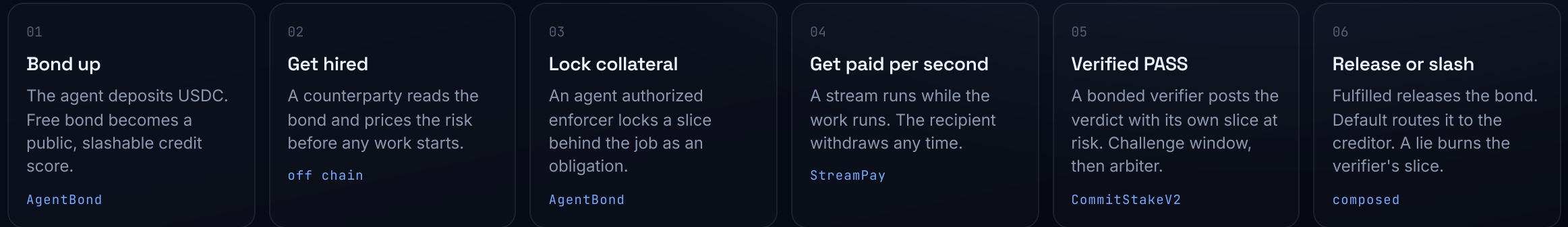
Not a Live badge over a mock. The anti collusion mechanism has fired on chain, twice.

WHAT	AMOUNT	ON CHAIN
Overturn burn, the surplus left after the harmed party is paid	1.45 USDC	0x97f31e7a...c45435 · Transfer to 0x...dEaD
Liveness burn, the whole slice when a dispute deadlocks	1.50 USDC	0x7bf59845...4aa6bd · Transfer to 0x...dEaD
Cross chain capital onboarding, Base Sepolia to Arc via CCTP V2	1 USDC	3 transactions, all status 1, bond 36 to 37
Source on arcscan, byte for byte with this repo	exact match	testnet.arcscan.app/address/0x1f1CA31b...

THE TESTS BITE, AND WE MEASURED IT RATHER THAN COUNTING THEM

A green number proves nothing on its own, so both fixes on the open branch were mutation checked: **removing the three new ceilings fails four tests, and reverting the helper to ignore the arbiter fee fails the regression test and the new fuzz property, which found its counterexample in two runs.** That is what says the suite would catch the bug coming back.

One agent lifecycle, settled end to end in USDC



Distinct wallets, not self dealing

Agent, creditor, enforcer, verifier and arbiter are separate addresses on chain. AgentBond #7 was released by the enforcer, #8 was slashed and the **creditor** received the 2 USDC. The agent cannot release or slash its own bond.

Ten terminal outcomes, all invariant tested

Fulfilled, defaulted, expired, overturned, deadlocked, abandoned. The solvency invariant holds across all of them: the contract can never owe more than it holds.

It plugs into the stack agents actually use

CIRCLE CCTP V2 · BRIDGE KIT

Capital does not have to start on Arc

1 USDC on Base Sepolia, bridged with Bridge Kit and deposited straight into AgentBond. Arc is already in the chain list, CCTP domain 26, so the cross chain leg is one `kit.bridge()` call. Three transactions, all status 1, runnable with `npm run onboard`.

ERC-8004

Real on chain identity, plus the money layer it lacks

Both demo agents hold identity NFTs in Arc's IdentityRegistry, with real third party feedback through ReputationRegistry and a separate execution wallet bound over EIP-712. ERC-8004 says who an agent is. It does not say what happens to the money when the agent lies. That is the gap we fill.

X402

Pay per call, settled per second underneath

A runnable demo gates an HTTP API behind a live StreamPay stream. The agent pays over HTTP 402 and the settlement is continuous rather than per invoice.

SDK · MCP SERVER

An agent can bond from inside its own tool loop

Single file ethers v6 wrapper with human USDC amounts, and an MCP server so any AI agent can check a passport, post a bond and open a bonded escrow, with quote before execute on every value moving call.

Not a prototype. A deployed, tested, documented stack.

4

contracts live on Arc testnet, ownerless,
source verified

187

Solidity tests: unit, adversarial, cold
audit, fuzz

5

Halmos symbolic proofs over the money
paths

150k

calls per invariant campaign (10k runs ×
depth 15), zero solvency violations

7

static dApp pages, no backend, live
RPC reads

3

runnable demos: commerce, x402,
CCTP

84.5%

measured mutation score, survivors
classified one by one

0

admin keys, proxies or pause switches

AND THE DOCUMENTATION A REVIEWER ACTUALLY NEEDS

THREAT_MODEL, SECURITY_AUDIT, VERIFIER_ECONOMICS, TEST_AUDIT, HALMOS_VERIFICATION, MUTATION_TESTING and a two minute JUDGES walkthrough. CI is green on every push and runs the same command a judge would.

What is still wrong with it, in our own words

Three findings are fixed on public branches and deliberately **not deployed**: changing a single character in a comment alters the solc metadata hash and breaks the live exact match verification. We measured that rather than assumed it.

FINDING	STATUS	BRANCH
Sockpuppet arbiter can reclaim the verifier's slice	Fixed and tested, opt in grieving guard	<code>fix/arbiter-griefing-optin</code>
AgentBond allowance race, revoke is not final	Fixed and tested, allowance epochs	<code>fix/agentbond-allowance-epoch</code>
Staker chosen time parameters were unbounded	Fixed and tested, three ceilings, 109 tests	<code>fix/commitstake-param-bounds</code>

Self correction, in public

Section 8 of the security document claimed the sockpuppet arbiter did not profit. It was wrong. We corrected it in the repo and on the page people actually read, rather than quietly deleting the claim.

THE LIMIT OF OUR OWN EVIDENCE

The invariant campaign bounds deadlines to thirty days, so a green campaign was never evidence against the unbounded lock finding. We wrote that into the public threat model as a limit of the thing we are otherwise proud of.

USDC as gas is what makes this economically possible

The chain is part of the design

- **One asset end to end.** Gas, bond, stream and slash are all USDC. An agent holds no volatile second token just to be able to act.
- **Micro settlement stays viable.** Per second streams and dust sized slices only make sense where fees are stable and denominated in the same unit as the value.
- **Neutral settlement between distrusting parties.** A shared database needs a trusted operator. These agents are not cooperating, so the ledger has to be.

Roadmap

- **Uncollateralized agent credit.** The headline use case: early agents borrow the bond against future cash flow. These primitives are the enforcement substrate that makes that recoverable.
- **Verifier set, not a verifier.** Stake backed set, dispute window, slashing for provably wrong attestations. The role is already an isolated address, so it is an interface change, not a redesign.
- **Enforcer accountability.** The same primitive turned on itself: the enforcer posts a bond and is slashable for a wrongful slash.
- **Redeploy and mainnet.** The three fix branches ship together, with re verification.

CHECK IT YOURSELF IN TWO MINUTES

Open a transaction. Run the suite. Nothing here needs your trust.

REPOSITORY

github.com/Mnorbert87/bondwire

LIVE DAPPS, NO BACKEND

mnorbert87.github.io/bondwire

BONDED VERIFIER FLOW, REAL TRANSACTIONS

mnorbert87.github.io/bondwire/bonded-verifier

TWO MINUTE JUDGE WALKTHROUGH

[JUDGES.md](#)

```
git clone https://github.com/Mnorbert87/bondwire
cd bondwire/contracts/commit-stake-v2 && forge test -vv
```